Pro- and Anti-Science Opinion Model (PASOM) Documentation

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1 Introduction and Purpose

This document describes the Pro- and Anti-Science Opinions Model (PASOM)—an agent-based model (ABM) of social media activity around a single science topic. The topic is one with a true (or likely true) belief (a pro-science belief) and a false (or likely false) belief (an anti-science belief), and agents update their beliefs over the probability of each position being true based on the actions of their connections. The documentation approximately follows *The ODD Protocol for Describing Agent-Based and Other Simulation Models* [1].

The model was designed to examine the potential role of toxic behaviour in social media interactions based on Spiral of Silence Theory predictions [2, 3]. In particular, the model incorporates sanctions and expected sanctions against opposition views [3], in the form of toxic behaviour. Toxic behaviour can cause agents' views to be silenced as they attempt to avoid toxic interactions with opposing agents. In turn, this may allow agents to be persuaded to change their mind.

Given the model's basis in the Spiral of Silence Theory, which explicitly includes majority and minority views, actions are divided into those in support of either the pro-science position or the anti-science position—agents cannot behave in a neutral or ambivalent manner. However, agents' views are not binary, so the model is a Continuous Beliefs Discrete Actions (CODA) model [4].

Outcomes of the model are reported in the paper, "Toxic behaviour facilitates echo chamber formation: An agent-based modelling simulation of science attitudes based on Spiral of Silence Theory" submitted to Plos One. Code to run the simulations used in the paper can be found on GitHub (https://github.com/timbainbridge/PASOM).

2 Agents, Variables, and the Network

2.1 Agents

Agents represent individuals in a single social media network. Each round, agents can choose to behave constructively, aiming to shift the opinions of all of their connections towards their side of the debate, or they can choose to behave toxically, aiming to reduce the likelihood that agents will share in support of the position opposing that of the toxic behaviour. Agents begin with a prior belief on the likelihood that the pro-science position is true and prior beliefs on the likelihoods that connected agents will behave constructively or toxically in support of or against both the pro- and anti-science positions.

Agents' decisions on whether to share or not are based on evaluations of payoff functions for sharing. Agents' payoffs are higher if they expect *constructive interactions*—connections' constructive behaviour matching their own—and lower if they expect *toxic interactions*—connections' toxic behaviour opposing the position they support. Agents therefore form expectations of the likelihood of each behavioural option of their connections. That is, they form expectations of their connections' toxic or constructive behaviour in support of the pro- and anti-science positions. These expectations are based on observations of past behaviour, inform their payoffs, and therefore affect their likelihood of performing each action. Payoff functions are also affected by a fixed cost of actions and randomness by round and randomness by round and agent.

If the values of one of their payoff functions is positive and larger than the other payoff function by a sufficient margin (determined by the u parameter), then the agent will post or share in support of the position matching the higher payoff.

Agents also evaluate utility functions to determine whether to form a new connection or sever an old connection. If they decide their utility would be greater with one or more extra connections, then they seek to form a new connection; if they think their payoff would be greater if the had one or more fewer connections, then they seek to sever a connection.

2.2 The Network

The starting network of agents can vary but must be undirected (i.e., representing Facebook, rather than Twitter/X) in the model's current form. The network is dynamic with agents seeking to connect to agents who started sharing chains that supported the position they supported and sever connections with those who acted toxically against the position they supported.

2.3 Variables

Agents are the primary unit of the model and nearly all variables include the subscript i, indicating that values differ by agent. The full set of variables are listed in Table 1. For a list of general terms and their definitions, which may be useful for interpreting variable definitions, see Table 2.

3 Process Overview

Each round (represented by j), the model proceeds through a number of steps (or submodels). The steps the model follows are:

1. Each agent has a q probability random chance of having the option to start a sharing chain. This represents agents who have some reason to share due to factors exogenous to the model (e.g., they see information about the topic from a source other than the social media network). If these agents choose to share in Step 2, they are *originator agents*.
2. All agents choose whether they would share in support of the pro- or anti-science position or not at all if they were part of a *sharing chain*. Sharing chains are created from originator agents through all other agents who choose to share and are connected, directly or indirectly, to an originator agent. Sharing chains can spread via those supporting opposing positions (consistent with [5]). The decision to share or not is based on evaluations of two payoff functions—one for the pro-science position and one for the anti-science position. If either is positive and the difference is greater than u, then they share in favour of the one with the higher payoff.
3. Agents who are part of a sharing chain decide whether to share constructively or toxically. If they share constructively, they share in favour of the position they decided to support in Step 2. If they share toxically, they decide to share in opposition to the position opposed to the one they decided to support in Step 2.
4. Agents use aggregations of shares with them to update their opinions, and their expectations of future constructive and toxic behaviour by their connections. They make separate assessments of likely constructive and toxic behaviour for both the pro- or anti-science positions. Agents also update their interest in the topic (see Section 7, Step 4).
5. Agents choose whether to disconnect from a connection who shared toxically against the position they supported that round. Inactive agents during the round do not disconnect. The decision to disconnect is based on assessments of whether their utility would increase with fewer agents supporting the relevant position.
6. Agents choose whether to create a new connection. New connections can only occur with originator agents who were in the same sharing chain as the agent (or if an originator themselves) and who shared in support of the same position. Connections are mutual as both agents must believe they would have higher utility with the extra connection (as for Facebook, given the undirected network).
7. Agents with no connections seek a like-minded connection. The connection is chosen randomly from candidates unless no options are available.

Each of these steps is described in detail in Section 7.

Table 1: Variables and their descriptions.

Variable	**Description**
ω_{ijI}	A binary variable indicating whether agent i was an originator agent for position I in round j.
e_{jI}	A normally distributed variable, with mean 0 and standard deviation of σ_e, indicating an exogenous bonus or penalty for position I for round j applied to all agents.
f_{ij}	A normally distributed variable, with mean 0 and standard deviation of σ_f, with a separate draw for each agent, i, and for each round, j. The variable represents separate exogenous motivation for each agent to engage each round.
U_{ijI}	The payoff of agent i for sharing in favour of position I in round j.
A_{ijI}	The number of constructive shares agent i observed for position I during round j.
T_{ijI}	The number of toxic shares agent i observed for position I during round j.
C_{iI}	A constant affecting agents' payoffs, representing a general cost based on time and effort for posting.
c_A	A variable affecting the payoff gained from observed constructive shares.
c_{AG}	A variable affecting the payoff gained from anticipated constructive shares from connections in general, rather than for sharing in a particular round (as for c_A). Used in the calculation for creating new connections or severing existing connections.
K_{ij}	The number of connections agent i has at round j.
K_{ijI}	Agent i's optimal number of connections for position I at round j.
π_{TijI} & π^*_{TijI}	The distribution of agent i's belief about the probability of a toxic response for supporting position I at round j from any single connection upon sharing. The '*' indicates the approximate median of the distribution.
α_{TijI}	A prior on agent i's perception of the likelihood of toxic responses following sharing in favour of position I in round j, defined as a decaying sum of the number of toxic shares seen by agent i opposing position I. One of the two parameters of agent i's prior on the probability of toxic responses along with K_{TijI}.
K_{TijI}	A prior on agent i's perception of the likelihood of toxic responses following sharing in favour of position I in round j, defined as a decaying sum of agent i's number of connections each round. The value can differ by position I due to differences in decay by I. One of the two parameters of agent i's prior on the probability of toxic responses for position I along with α_{TijI}.
π_{AijI} & π^*_{AijI}	The distribution of agent i's belief about the probability of a constructive response for position I at round j from any single connection upon sharing. The '*' indicates the approximate median of the distribution.
α_{AijI}	A prior on agent i's perception of the likelihood of constructive responses following sharing in favour of position I in round j, defined as a decaying sum of the number of constructive shares seen by agent i supporting position I. One of the two parameters of agent i's prior on the probability of constructive responses along with K_{AijI}.
K_{AijI}	A prior on agent i's perception of the likelihood of constructive responses following sharing in favour of position I in round j, defined as a decaying sum of agent i's number of connections each round. The value can differ by position I due to differences in decay by I. One of the two parameters of agent i's prior on the probability of constructive responses for position I along with α_{AijI}.
λ_{ij}	A multiplier for loss of interest for agent i for round j. When interested, equal to 1, otherwise equal to ν.
p_{Xij}	The probability agent i will share toxically in round j (given agent i has decided to share, else NA).
p_{Sij}	Not to be confused with π_{Sij}, p_{Sij} is the proportion of shares seen by agent i in round j that were pro-science (either toxic or constructive).
α_{Sij}	The sum of all pro-science shares with agent i from round 1 to round j, multiplied by the pro-science bias parameter, θ (see Table 3), plus the starting value, α_{Si0}. One of the parameters of the prior of opinion along with β_{Sij}.
β_{Sij}	The sum of all anti-science shares with agent i from round 1 to round j, divided by the pro-science bias parameter, θ (see Table 3), plus the starting value, β_{Si0}. One of the parameters of the prior of opinion along with α_{Sij}.
π_{Sij} & π^*_{Sij}	The probability distribution of agent i's science belief at round j. The '*' indicates the approximate median of the distribution.

Table 2: General model terms and subscripts, and their descriptions.

Term	Description
I	The side of the issue. 1 = Pro-science; 0 = Anti-science.
j	The round number. I.e., $j = 1$ is the first round. $j = 0$ may represent staring conditions in some cases.
i	The agent ID.
N	The number of agents.
A	Used to indicate either the number of constructive (agreeable or positive) shares or, as a subscript, a related parameter.
T	Used to indicate toxic shares seen or, as a subscript, a related parameter.
C	Used to indicate the cost of sharing or, as a subscript, a related parameter.
G	Added to A, T, C, B, or K to indicate a parameter or variable that applies to the 'general' utility function (see Equation (18)), rather than the payoff function for action.
K	Used to indicate something related to the number of connections agents have.
X	Used to indicate toxic behaviour or a related parameter.
S	Used to indicate a parameter related to science belief or opinion.

4 Design

4.1 General Design Considerations

The overall purpose of creating the model was to evaluate how mechanisms described by Spiral of Silence Theory [2, 6] might play out in an online social media environment with toxic behaviour acting as the expected sanction [3]. Specifically, with the model, we aimed to test the theoretical prediction that toxic behaviour could be an important mechanism behind echo chamber formation in a model of the theorised elements. Therefore, the primary consideration for most of the design of the model was how to incorporate toxic behaviour in such a way that it would affect agents in the desired manner. This led us to select a CODA model since discrete actions allowed agents to evaluate only two payoff functions, while also providing an obvious way for toxic behaviour to affect those payoffs—that is, toxic behaviour against a position would reduce payoffs for the opposing position in subsequent rounds. Had ambivalent activity been allowed, we would have had to specify payoff functions for this activity and how it should be affected by toxic behaviour at the extremes. It was not clear to us how either of these should be done. Moreover, as noted above, a CODA model seemed the most appropriate to capture Spiral of Silence considerations, given its focus on the binary framing of majority versus minority views.

Once we had decided upon a CODA model, we chose to make the model about an arbitrary science topic, where agents could support either a pro- or anti-science position. This was done because many science topics can be narrowed down to single issues such that a binary set of opinions is reasonable. For example, the option to get vaccinated against COVID-19 can be considered as a binary choice that one can support or oppose. Of course, this is a gross simplification but one that may be sufficiently close to reality and matches the Spiral of Silence framing.

A second consideration for using a pro- vs anti-science conceptualisation is that it allowed us to uncontroversially place a bias in the model towards the pro-science view, given that most people hold a pro-science view on most topics [7, 8, 9]. This was important as Spiral of Silence Theory includes majority and minority views, so there needed to be a way to make one view typically end up as the majority view so that the Spiral of Silence Theory prediction that minority views are suppressed by expected sanctions (in this case toxic behaviour) could be evaluated. Our model could also consider a generic topic where support was binary and opinions were biased towards one side of the debate, however, the pro- versus anti-science framing is not inappropriate and motivated our research.

4.2 Emergence

Emergent outcomes of simulations are the network structure, the distribution of opinions, and echo chamber membership. Each of these emergent features is influenced by the implementation of agents' behaviour through payoff functions and their assessments of the likelihoods of connections' actions based on past behaviour. These actions can cause agents with differing views to either disconnect or for one or other view to be suppressed such that agents can be persuaded to change their view. Randomness in the payoff functions means the outcome is not assured and, if agents opinion priors are relatively weak, agents can adopt either position making the simulation outcomes unpredictable.

Nevertheless, unless toxic behaviour is set to occur sufficiently rarely, toxic behaviour in the model means that agents with different views will either disconnect or suppress their opinion. Unless they have a very strong belief, they will then be persuaded. This means echo chambers are highly likely to form if agents start with or develop extreme opinions (even if weak and easy to persuade) and randomness and the discrete actions model makes it likely that will occur.

4.3 Adaptation

There are three main submodels where agents behaviour is adaptive. First, agents' decisions on whether to share or not adapts to their connections' actions in prior rounds. Agents tend to share more frequently when they expect constructive interactions and less frequently when they expect toxic interactions, reflecting reasonable expectations of how most people act and consistent with mechanisms from the Spiral of Silence Theory. Second, agents' decisions on whether to share constructively or toxically adapt to activity in the current round. Agents will not act toxically when not faced with an opposing opinion and their probability of acting toxically increases as the proportion of opposition interactions increases (to a theoretical maximum of 1, depending on other variable and parameter values, see Equation (7)).[1] Third, agents adapt to their expectations of constructive and toxic interactions in their choices on whether to sever existing connections and form new connections. Expectations of many constructive interactions increase agents' optimal number of connections, making them more likely to form new connections, whereas expectations of many toxic interactions decrease agents' optimal number of connections, making them more likely to sever existing connections.

4.4 Objectives

Agents objectives are implicit in their sharing behaviour—they aim to experience constructive interactions and avoid toxic interactions, while persuading other agents to their view or suppressing the opposing view. These objectives are reflected in their payoff functions for sharing and also in their probability of sharing toxically once they have decided to share. If agents always shared constructively, then they may not be able to suppress and overcome an opposition position in their segment of the network. On the other hand, by periodically behaving toxically, agents can suppress the activity of support for the opposition, leaving them free to persuade agents to their opinion in subsequent rounds.

In their connection decisions, agents' objectives are explicit—they aim to maximise their utility by severing toxic connections and forming connections expected to be constructive (see Equation (18)).

4.5 Learning

Agents learn based on the actions of their connections. They learn what behaviour to expect in the form of the likelihood of constructive and toxic behaviour in support of either the pro- or anti-science

[1] While we chose to make the probability of toxic behaviour strictly increasing with the proportion of shares opposed to an agent's view, it is not clear how this function should have been implemented given the Spiral of Silence Theory's focus on minority view suppression and not the actions of minority view holders when they choose to act.

position, and they learn what to believe on the science topic. These expectations and their opinions are updated using Bayesian reasoning, described in Section 7 (see Equations (9), (10), (12), (13), (15), and (16)). Bayesian reasoning was selected as it is has a mathematical basis so could readily be applied in the model, and it is plausibly a reasonable approximation of human cognition [10], including some cognitive biases [11], although not all agree [12].

4.6 Prediction

Agents predict the number of constructive and toxic interactions they would have were they to support each view (see Equations (2) and (3)). When choosing to form or sever connections, agents also assume that the change will increase or decrease their utility (see Equation (18)).

4.7 Sensing

Agents know the aggregation of how their connections share each round. That is, they know how many of their connections behaved constructively in support of both the pro- and anti-science positions, and they know how many of their connections behaved toxically in opposition to the pro- and anti-science positions. Agents also know which agents are originator agents in the same sharing chain as themselves who shared in support of the same position. Finally, agents with fewer than two connections after the main connection step know which agents have an opinion within γ or on the same side of the issue as their own.

4.8 Interaction

All interactions are direct. When agents choose to share all their connections see their behaviour and no other agents see their behaviour. The one exception is for new connection formation, when agents can connect with an originator agent in the same sharing chain who shared in support of the same position.

4.9 Stochasticity

The main random elements are random bonuses or penalties to support each position. One of these is a general bonus or penalty, applied separately for each position to all agents for the round (i.e., e_{jI}, see Tables 1 and 2), the other is an individual bonus or penalty, applied to both positions, with a separate draw for each agent, each round (i.e., f_{ij}, see Tables 1 and 2). These were included for two reasons. First, pragmatically, without these variables, agents would be deterministic, given expectations. Second, it is reasonable to assume that different news or other things happening in the world might affect all agents' desire to share in support of a particular position or that particular events unique to individuals might affect their desire to engage with the topic from round to round. In reality, these may be auto-correlated (i.e., an event on day one could boost engagement in subsequent days, independent of others' actions) but this has not been modelled in the current version.

The other random element is whether agents choose to act toxically or not. However, given a certain value of the key parameter, c_X (see Table 3), it is unlikely this has much affect on model outcomes as the avoidance of toxic behaviour in one round means that the support of the opposing position is not silenced and the motivation to act toxically in subsequent rounds remains. This is discussed in the paper (noted above) with respect to the model's insensitivity to changes in c_X for higher values of c_X.

4.10 Collective

Echo chambers form as an emergent property of the network and model dynamics but are at no stage modelled explicitly. By definition, they, or rather agents within them, interact with agents with an opposing view relatively infrequently.

Table 3: Model parameters and their descriptions.

Parameter	Description
q	The probability that an agent is an originator agent in round j.
σ_e	The standard deviation of e_{jI}.
σ_f	The standard deviation of f_{ij}.
c_T	A parameter affecting the payoff lost from toxic interactions.
c_{TG}	A parameter affecting the utility lost from anticipated toxic interactions from connections in general, rather than for sharing in a particular round (as for c_T).
c_C	A parameter affecting the utility lost from interacting. Simplifies to be the cost of interacting (excluding toxic interactions).
b_A	A parameter affecting the utility gained from constructive interactions via c_A.
b_{AG}	A parameter affecting the utility gained from anticipated constructive interactions from connections in general, rather than for sharing in a particular round (as for b_A).
c_{BG}	A parameter defining the benefit per connection of maintaining connections in general.
c_{KG}	A parameter defining the cost per connection of maintaining the connection in general.
u	A parameter determining the payoff differential required to post or share one view over the other.
μ	A parameter affecting how quickly agents lose interest.
ν	A parameter indicating how much slower agents update their belief after losing interest.
c_X	A parameter affecting agents' propensity to share toxically.
α_{Si0}	One of two vectors representing agents' initial science belief priors, along with β_{Si0}. This is a vector (with a different value for each agent i) because agents start with different priors on science beliefs.
β_{Si0}	One of two vectors representing agents' initial science belief priors, along with α_{Si0}. This is a vector (with a different value for each agent i) because agents start with different priors on science beliefs.
α_{T0}	One of two values representing agents' initial toxic behaviour expectations prior, along with K_0. It is a single value as agents' initial expectations do not differ.
α_{A0}	One of two values representing agents' initial constructive behaviour expectations prior, along with K_0. It is a single value as agents' initial expectations do not differ.
K_0	One of two values representing both agents' initial toxic behaviour expectations prior and their initial constructive behaviour expectations, as the same values are used for this parameter for both expectations for all agents.
d_T	The decay rate of toxic interactions, with 1 being no decay and 0 indicating completely myopic agents.
d_A	The decay rate of constructive interactions, with 1 being no decay and 0 indicating completely myopic agents.
θ	A weighing multiplier favouring pro-science shares and penalising anti-science share if > 1 or favouring anti-science shares and penalising pro-science share if < 1. Neutral if equal to 1.
γ	A parameter representing tolerance for opposing views when connecting to a random agent in Step 7.

4.11 Observation

Each round (j), values are saved for every agent (i) in $i \times j$ matrices. These are described in Table 4. Additionally, the original and final networks, and agents' final posteriors for opinion, and constructive and toxic behaviour for both values of I are saved. The final posteriors are saved in matrix of values for agents that is used throughout the simulations to track agents' values on all multiple variables, including priors / posteriors, π^*_{Sij}, and λ_{ij}.

Table 4: Observed outcomes.

Symbol	Description	Possible Values
π^*_{Sij}	The median of agent i's opinion distribution in round j (see Equation (17)).	$(0, 1)$
K_{ij}	Agent i's number of connections in round j.	$\{0, N-1\}$
A_{ij1}	Whether agent i shared constructively in favour of the pro-science position in round j.	$0, 1$
A_{ij0}	Whether agent i shared constructively in favour of the anti-science position in round j.	$0, 1$
T_{ij1}	Whether agent i shared toxically in opposition to the pro-science position in round j.	$0, 1$
T_{ij0}	Whether agent i shared toxically in opposition to the anti-science position in round j.	$0, 1$
ω_{ij1}	Whether agent i was an originator agent for the pro-science position in round j.	$0, 1$
ω_{ij0}	Whether agent i was an originator agent for the anti-science position in round j.	$0, 1$
λ_{ij}	Agent i's interest in the topic in round j (see Equation (5)).	$\nu, 1$

5 Initialisation

Agents begin with priors for their belief on the science topic and their expectations of constructive and toxic behaviour for pro- and anti-science behaviour. Each of these beliefs is represented by α and β values in a beta distribution. In versions run for the first paper to use the model (noted above), science belief priors were distributed in a bell-curve shape across most of the 0 to 1 range and were generated with the aid of random numbers drawn from a normal distribution. These priors were created such that agents with more extreme views had stronger views. The strength differed for different simulations. Priors for constructive and toxic behaviour were relatively weak and identical across agents to enable actions to reflect activity in the model quickly.

Similarly, agents begin with expectations of the probability of constructive and toxic interactions. In versions run for the paper, these were set to be equal for all agents and across both positions.

Finally, a starting network is required to specifies which agents are connected to each other. In simulations run for the paper, these were either generated randomly in a cluster structure or, in one case, a lattice network with agents placed randomly within the lattice network.

6 Input Data

No input data is used throughout the running of the model, beyond the initial setup. Values of e_{jI} could be set and used as input data (rather than generated randomly) to reflect a particular topic over a particular time frame.

7 Detailed Steps (Submodels)

This section presents the details of the steps introduced in Section 3 above. The complete collection of terms, parameters, and variables used in the equations presented in this sections are listed in Tables 1 (variables), 2 (terms), and 3 (parameters), respectively.

7.1 Step 1

Every round, random draws from a Bernoulli distribution with probability q are made for all agents to determine which agents have the option to start a sharing chain during the round, thereby becoming originator agents. A sharing chain is a subnetwork of the full network including at least one originator agent and any agents who choose to share and who are connected, directly or indirectly, to the originator agent(s) (see Step 2). All agents' probability of being selected as a possible originator agent is independent of all other agents' selections and draws are independent across rounds. From this selection, any agent who decides to share (see Step 2) is an originator agent, indicated by ω_{ijI} for agent i in round j for the position I they decided to share in favour of.

7.2 Step 2

Agents decide whether or not to share for the round. This is decided by all agents simultaneously and independently. Agents who had the option of becoming originator agents in Step 1 become originator agents if they decide to share. For all other agents, only those who are connected to an originator agent via a sharing chain will actually be able to share. Those who would have shared but are not connected to an originator agent by a sharing chain, do not share. This is done because it is much quicker than iterating through connections to create the set of agents who share.

Agents can decide whether to share with a pro- or anti-science message or not at all. To make their decision, agents perform two payoff (or utility) calculations. One calculation evaluates the net benefit of sharing in support of the pro-science position and the other evaluates the net benefit of sharing in support of the anti-science position. If one of these values is positive, then agents share in support of the position matching the higher value. If neither value is positive, then agents do not share.

For each position I, the payoff, U_{ijI}, is a function of the expected number of constructive interactions, $\mathbb{E}[A_{ijI}]$; the expected number of toxic interactions, $\mathbb{E}[T_{ijI}]$; the expected cost of interacting (e.g., time spent, effort expended), $\mathbb{E}[C_{iI}]$; an exogenous bonus or penalty for the position and round, e_{jI}; and an exogenous bonus or penalty for each agent for the round, f_{ij}. e_{jI} is normally distributed with a mean of 0 and a standard deviation of σ_e, and is independent across rounds and position. f_{ijI} is normally distributed with a mean of 0 and a standard deviation of σ_f and, as for e_{jI}, is independent across rounds, position, and, additionally, individuals. Thus,

$$U_{ijI} = F(\mathbb{E}[A_{ijI}],\ \mathbb{E}[T_{ijI}],\ \mathbb{E}[C_{iI}],\ e_{jI},\ f_{ij})$$

For simplicity, each value is assumed to be separable,

$$U_{ijI} = F_1(\mathbb{E}[A_{ijI}]) + F_2(\mathbb{E}[T_{ijI}]) + F_3(\mathbb{E}[C_{iI}]) + F_4(e_{jI}) + F_5(f_{ij})$$

Despite the i and I subscripts, C_{iI} is assumed to be the same for all agents, the same over all rounds, and the same for each position, and is known to all agents. Therefore, $\mathbb{E}[C_{iI}] = C$. These subscripts were included to indicate that costs could, in theory, differ by agent or position, although they do not here. With the exception of constructive interactions, A_{ijI}, for all other variables, a one unit increase in the expectation leads to a constant change in payoff, regardless of the value of the payoff or the variable in question. For $F_1(\mathbb{E}[A_{ijI}])$, the payoff due to each expected constructive interaction follows a natural log function with a utility of 0 for no expected interactions; that is, $F_1(\mathbb{E}[A_{ijI}]) \propto \ln(\mathbb{E}[A_{ijI}]+1)$. Thus,

$$U_{ijI} = c_A \ln\left(\mathbb{E}[A_{ijI}] + 1\right) - c_T\, \mathbb{E}[T_{ijI}] - c_C C + f_{ij} + e_{jI} \tag{1}$$

where c_A, c_T, and c_C set the effects of constructive interactions, toxic interactions, and costs, respectively. Given C and c_C are constants, for convenience, C is set to 1. Thus, $c_C C = c_C$ for all agents.

Agents use Bayesian updating [13] to inform their expectations of the number of constructive and toxic shares they are likely to see if they share. These are represented by A (agreeable) for constructive and T for toxic. The process for these updates are described in detail in Step 4 where agents update their expectations for the following round. For now, it is enough to state that agents' expected number of toxic interactions is the belief about the probability that any one agent will act toxically multiplied by their number of connections. Thus,

$$\mathbb{E}[T_{ijI}] = K_{ij}\pi^*_{TijI} \tag{2}$$

and,

$$\mathbb{E}[A_{ijI}] = K_{ij}\pi^*_{AijI} \tag{3}$$

where π^*_{TijI} is agent i's estimation of the probability that any one connection will behave toxically in support of I in round j and π^*_{AijI} is agent i's estimation of the probability that any one connection will behave constructively in support of position I in round j.

Putting these values into the payoff functions requires one more step. For toxic interactions, the payoff is a simple multiplier of the expected number of constructive shares. However, the payoff calculation for constructive shares also includes a parameter indicating the median of the probability one thinks the position is true, π^*_{Sij} (see Step 4 for further details) and a variable indicating when an agent loses interest in the topic, λ_{ij}. This variable is added to the equation because agents get less benefit from constructive interactions if they are no longer interested in the topic or if they disagree with the shared view.

Thus,

$$c_A = \begin{cases} b_A\lambda_{ij}\pi^*_{Sij} & \text{for } I = 1 \\ b_A\lambda_{ij}(1-\pi^*_{Sij}) & \text{for } I = 0 \end{cases} \tag{4}$$

where b_A is a parameter representing the benefit accrued from positive interactions following sharing, and λ_{ij} (loss of interest) is defined as follows:

$$\lambda_{ij} = \begin{cases} \nu & \text{if } (\alpha_{Sij} + \beta_{Sij})|\pi^*_{Sij} - 0.5| > \mu \\ 1 & \text{if } (\alpha_{Sij} + \beta_{Sij})|\pi^*_{Sij} - 0.5| \le \mu \end{cases} \tag{5}$$

The calculation of λ_{ij} depends on how likely the agent thinks one side or the other is true, $|\pi^*_{Sij} - 0.5|$; the sum of the variables defining their priors for the current round, $\alpha_{Sij} + \beta_{Sij}$; a parameter indicating how much agents' updating decreases once they lose interest, ν (e.g., a value of 0.1 means new observed shares would be weighted at 10% the weight of prior shares); and a parameter indicating how strong and extreme an opinion needs to be for an agent to lose interest, μ. The term $\alpha_{Sij} + \beta_{Sij}$ is the total of the shares agent i has seen combined with the original values of these variables (see Step 4 for further information). Thus, for a given number of shares seen, agents will lose interest earlier if they have a stronger opinion (one way or the other), whereas they will remain interested for longer if they are still undecided on the correct view. When exactly agents lose interest, given the other values, is defined by μ. (For example, if an agent had a very strong pro-science view of $\pi^*_{Sij} = .95$, then, for $\mu = 1000$, they would become disinterested at $\alpha_{Sij} + \beta_{Sij} = 2,222$.)

Thus, from Equations (1)–(4),

$$U_{ijI} = \begin{cases} b_A \lambda_{ij} \pi^*_{Sij} \ln\left(K_{ij} \pi^*_{Aij1} + 1\right) - c_T K_{ij} \pi^*_{Tij1} - c_C + f_{ij} + e_{j1} & \text{for } I = 1 \\ b_A \lambda_{ij} (1 - \pi^*_{Sij}) \ln\left(K_{ij} \pi^*_{Aij0} + 1\right) - c_T K_{ij} \pi^*_{Tij0} - c_C + f_{ij} + e_{j0} & \text{for } I = 0 \end{cases} \tag{6}$$

with values within determined by Equations (5), (11), and (14).

From (6), if $U_{ijI} < 0$, for both $I = 1$ and $I = 0$, then agent i will not share. If $U_{ijI} > 0$ for $I = 1$ or $I = 0$ and $|U_{ij1} - U_{ij0}| > u$, then agent i will share in support of the option with the higher expected payoff. Given the presence of f_{ij} and e_{jI} in the equation, with all other variables held constant, different decisions could be reached in marginal cases (depending on the value of u, and σ_e and σ_f–the standard deviations of f_{ij} and e_{jI}).

7.3 Step 3

Once agents have decided to share, and which polarity their share should take, they decide whether to share constructively or toxically. The decision to share toxically is based on the idea that creating an unpleasant environment to share views opposing one's own can be an effective method of silencing opposing ideas, thereby reducing the idea's influence. However, agents also want to maintain the chance of persuading agents who may disconnect from them if they behave toxically.

The probability of sharing toxically, p_{Xij}, is,

$$p_{Xij} = \begin{cases} \min(1, 2c_X(\pi^*_{Sij} - 0.5)(1 - p_{Sij})) & \text{for } \pi^*_{Sij} > 0.5 \ \& \ I = 1 \\ \min(1, 2c_X(0.5 - \pi^*_{Sij}) p_{Sij}) & \text{for } \pi^*_{Sij} < 0.5 \ \& \ I = 0 \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

where X indicates a variable related to toxic sharing (as opposed to T, representing seeing toxic shares), c_X is a parameter affecting the likelihood of toxic interactions, and p_{Sij} is the proportion of shares supporting the pro-science position among agent i's contacts during round j.

The proportion of shares supporting the pro-science position among agent i's contacts during the round, p_{Sij}, is calculated from the total activity of agent i's connections during the round[2],

$$p_{Sij} = \frac{(A_{ij1} + T_{ij0})}{(A_{ij1} + A_{ij0} + T_{ij1} + T_{ij0})} \tag{8}$$

where A_{ijI} is the number of constructive shares supporting position I, in round j, seen by agent i; and T_{ijI} is the number of toxic shares opposing position I, in round j, seen by agent i. Therefore, $A_{ij1} + T_{ij0}$ is the number of pro-science shares seen by agent i in round j, $A_{ij1} + A_{ij0} + T_{ij1} + T_{ij0}$ is the total number of shares seen by agent i in round j, and p_{Sij} is the proportion of shares seen by agent i in round j that are pro-science.

Thus, from Equation (7), if all the behaviour observed by a particular agent is for the view they agree with, then the agent will share constructively, and, as the proportion moves in favour of the idea opposed to their own, their chance of persuading their connections in an open environment decreases and their

[2]This calculation is possible because all agents act simultaneously during each round, rather than iteratively. Arguably, this could be based only on the behaviour of connections closer to an originator agent, but the added complexity did not warrant the marginal benefit the results, so it was not done.

chance of sharing toxically increases. This increase is moderated by the extremity of their view, π^*_{Sij}, such that those with stronger views are more likely to act toxically in the presence of opposition. If $c_X = 1$, then p_{Xij} would be constrained to the range $[0, 1]$ without taking a minimum; however, with $c_X > 1$, p_{Xij} could exceed 1 if not constrained. It was therefore constrained to a maximum of 1 as probabilities cannot exceed 1.

7.4 Step 4

After sharing, agents update their beliefs with Bayesian reasoning [4, 13]. These beliefs include their beliefs about the likely probability of connections sharing constructively (π_{AijI}) and toxically (π_{TijI}) and their opinion about the probability of the pro-science position being true (π_{SijI}).

From Bayes Rule, we know,

$$P(\pi_{TijI}|\alpha_{TijI}, K_{TijI}) \propto P(\alpha_{TijI}|K_{TijI}, \pi_{TijI})P(\pi_{TijI}|K_{TijI})$$

where T indicates that each value relates to toxic interactions; α_{TijI} is the number of toxic shares agent i has seen for position I; K_{TijI} is the sum of the number of contacts each round (see below); and π_{TijI} is the probability that a contact will share toxically if agent i posts in support of position I in round j.

Typically, probability in a Bayesian context is not a function of anything, but represents the probability of an unchanging fact about the world (e.g., probability that a coin lands on heads when tossed). However, here, π_{TijI} is a function of agent i's connections so is different for different agents and can change over time. This is reflected in both K_{TijI} and α_{TijI}, which are weighted towards recent interactions to reflect possible changes to π_{TijI} over time. Moreover, although incorrect, agents assume π_{TijI} is equal for all connections—a simplifying assumption that makes the mathematics much easier—and that decisions to share toxically are independent[3].

Given these assumptions,

$$\alpha_{TijI}|K_{TijI}, \pi_{TijI} \;\sim\; \text{Bin}(K_{TijI},\; \pi_{TijI})$$

Assuming a beta distributed prior for $P(\pi_{TijI}|K_{TijI})$, agent i's estimation of π_{TijI} is also a beta distribution. That is,

$$\pi_{TijI} \;\sim\; \text{Beta}(\alpha_{TijI}, K_{TijI} - \alpha_{TijI})$$

Given that agents realise the characteristics of the system change, α_{TijI} and K_{TijI} decay over time with each value being calculated as the sum of a proportion of prior observations plus new observations. Thus,

$$\alpha_{TijI} = d_T\alpha_{Ti(j-1)I} + T_{ijI} \tag{9}$$

and,

$$K_{TijI} = d_T K_{Ti(j-1)I} + K_{ij} \tag{10}$$

[3]Although toxic share decisions are not explicitly dependent, the decisions of agents' connections to share toxically are a function of their connections' behaviour. If agents are more likely to be connected to connections' connections (which they likely are due to initial network setup and the way new connections form) and they are likely to share similar opinions (also true, due to belief updating), then agents' assumptions that toxic shares by connections are independent is also likely not strictly true.

where α_{TijI} is the decaying number of toxic shares seen by agent i for position I up to and including round j; $\alpha_{Ti(j-1)I}$ is the value of α_{TijI} for the $(j-1)$ round (where round j is the current round); K_{TijI} is the decaying sum of number of contacts agent i has over all rounds up to the current round; $K_{Ti(j-1)I}$ is value of K_{TijI} for the round $(j-1)$; d_T is the decay rate of toxic shares; and T_{ijI} is the number of toxic shares seen by agent i opposing position I in round j. The initial priors at round $j = 0$ are α_{T0} and K_0 (which do not differ by i nor I, see Table 3).

To create a single value of each agent's belief, π^*_{TijI} is calculated as the median of the distribution. Thus,

$$\pi^*_{TijI} \approx \frac{\alpha_{TijI} - \frac{1}{3}}{K_{TijI} - \frac{2}{3}} \tag{11}$$

The median of a beta distribution is only defined this way for $\alpha_{TijI} \geq 1$ and $K_{TijI} - \alpha_{TijI} \geq 1$. Because these values can decay, they could fall below 1 if few toxic share occur. Therefore, $d_T = 1$ whenever the value of α_{TijI} (or K_{TijI}) would fall below 1. This is also why K_{Tij1} is distinguished from K_{Tij0} as, without this constraint, the decaying number of connections would always be the same given the same decay rate.

Expected benefits of constructive interactions are calculated similarly. That is,

$$\alpha_{AijI}|K_{AijI}, \pi_{AijI} \;\sim\; \text{Bin}(K_{AijI},\; \pi_{AijI})$$

and,

$$\pi_{AijI} \;\sim\; \text{Beta}(\alpha_{AijI}, K_{AijI} - \alpha_{AijI})$$

with,

$$\alpha_{AijI} = d_A \alpha_{Ai(j-1)I} + A_{ijI} \tag{12}$$

and,

$$K_{AijI} = d_A K_{Ai(j-1)I} + K_{ij} \tag{13}$$

All variables are defined as for toxic shares with constructive shares replacing toxic ones. The initial priors are α_{A0} and K_0. Note that, because of different decay parameters for constructive and toxic shares and the same constraint keeping $\alpha_{AijI} \geq 1$, K_{Aij1} need not equal K_{Aij0} and K_{AijI} need not equal K_{TijI} for either value of I.

Taking π^*_{AijI} as the median of the beta distribution gives,

$$\pi^*_{AijI} \approx \frac{\alpha_{AijI} - \frac{1}{3}}{K_{AijI} - \frac{2}{3}} \tag{14}$$

Agents also use Bayesian reasoning to update their opinion on the science topic. When doing so, agents assume that the information they receive, in the form of constructive shares by connections, is unbiased. They are also not good at evaluating the truth-value of information, such that receiving a lot of anti-science information will make them hold an anti-science belief (depending on the value of θ, see Table 3).

From Bayes Rule, we know that,

$$P(\pi_S|\alpha_{Sij}, \beta_{Sij}) \propto P(\alpha_{Sij}|K_{Sij}, \pi_S)P(\pi_S|K_{Sij})$$

where α_{Sij} is the total number of pro-science shares agent i has seen, β_{Sij} is the total number of anti-science shares one has seen, $K_{Sij} = \alpha_{Sij} + \beta_{Sij}$, and π_S is the probability that the pro-science position is 'true'.

However, if the pro-science position is true for a particular topic, then unbiased information would always or nearly always indicate the truth of the pro-science information and agents' opinions would tend towards a belief of 1. Even with some chance of false negatives, unbiased information should overwhelmingly support the pro-science position (and vice versa when the scientific consensus is wrong).

However, the information seen by agents is filtered through the beliefs of other agents and, therefore, it is not unbiased. The bias is such that, if many of an agent's connections hold an anti-science position, then the agent's belief may also tend in that direction. Thus, agents' Bayesian updating is not based on the unbiased π_S but is instead based on the biased π_{Sij}, which is agent i's perception of π_S given the information shared by their connections up to round j. However, agents are biased towards the pro-science side, given their connections' views, represented by θ (as long as $\theta > 1$), which increases agents' weighting of pro-science shares and decreases their weighting of anti-science shares.

Adding these changes and treating each piece of information as independent,

$$\alpha_{Sij}|K_{Sij}, \pi_{Sij} \sim \text{Bin}(K_{Sij}, \pi_{Sij})$$

Assuming π_{Sij} is independent of K_{Sij}, the posterior after j time periods is,

$$\pi_{Sij} \sim \text{Beta}(\alpha_{Sij}, \beta_{Sij})$$

where,

$$\alpha_{Sij} = \alpha_{Si(j-1)} + \lambda_{ij}A_{ij1}\theta \tag{15}$$

and,

$$\beta_{Sij} = \beta_{Si(j-1)} + \frac{\lambda_{ij}A_{ij0}}{\theta} \tag{16}$$

The initial prior at $j = 0$ is set by the parameters α_{Si0} and β_{Si0}, which differ by i as each agent can have different sets of priors on the topic. The λ_{ij} variable is included because agents update more slowly when they have lost interest in the topic.

To get opinion as a single number (rather than a distribution), take the median,

$$\pi^*_{Sij} \approx \frac{\alpha_{Sij} - \frac{1}{3}}{\alpha_{Sij} + \beta_{Sij} - \frac{2}{3}} \tag{17}$$

7.5 Step 5

To determine disconnections and new connections, a payoff function, similar to the payoff function from earlier (Equation (6)), represents the expected net benefit of being part of the network, rather than of sharing. Unlike in the case of sharing, agents gain a small, diminishing benefit from having connections and face a small cost for maintaining connections.[4]

$$V_{ijI} = c_{AG} \ln(\mathbb{E}[A_{i(j+1)I}] + c_{BG}K_{i(j+1)} + 1) - c_{TG}\,\mathbb{E}[T_{i(j+1)I}] - c_{KG}K_{i(j+1)} \tag{18}$$

where V_{ijI} is the payoff of being part of the network; c_{AG} is the expected payoff from constructive interactions in general; c_{TG} is the perceived cost per toxic interactions in general; c_{BG} is a benefit per connection of maintaining connections; and c_{KG} is a cost per connection of maintaining the connection. The equations are prospective, with agents considering their number of connections in the subsequent round ($K_{i(j+1)}$), and the expected number of toxic ($\mathbb{E}[T_{i(j+1)I}]$) and agreeable ($\mathbb{E}[T_{i(j+1)I}]$) interactions in the subsequent round.

For simplicity, agents assume that the rate of toxic and agreeable interactions will remain the same in future rounds, thus, given Equations (2) and (3),

$$V_{ijI} = c_{AG} \ln\left(K_{i(j+1)}\pi^*_{AijI} + c_{BG}K_{i(j+1)} + 1\right) - c_{TG}K_{i(j+1)}\pi^*_{TijI} - c_{KG}K_{i(j+1)} \tag{19}$$

The derivative of V_{ijI} with respect to $K_{i(j+1)}$ is,

$$\frac{\partial V_{ijI}}{\partial K_{i(j+1)}} = \frac{c_{AG}(\pi^*_{AijI} + c_{BG})}{K_{i(j+1)}(\pi^*_{AijI} + c_{BG}) + 1} - c_{TG}\pi^*_{TijI} - c_{KG}\,. \tag{20}$$

V_{ijI} reaches a maximum at $\frac{\partial V_{ijI}}{\partial K_{i(j+1)}} = 0$. However, the value of $K_{i(j+1)}$ where this maximum will be reached varies depending on I. Therefore, V_{ijI} reaches a maximum when,

$$0 = \frac{c_{AG}(\pi^*_{AijI} + c_{BG})}{K_{i(j+1)I}(\pi^*_{AijI} + c_{BG}) + 1} - c_{TG}\pi^*_{TijI} - c_{KG}\,,$$

with $K_{i(j+1)I}$ indicating that different values of $K_{i(j+1)}$ are desired for the pro- versus anti-science positions.

Rearranging,

$$K_{ijI} = \frac{c_{AG}}{c_{TG}\pi^*_{TijI} + c_{KG}} - \frac{1}{\pi^*_{AijI} + c_{BG}} \tag{21}$$

where c_{AG} varies similarly to c_A (as specified in Equation (4)), that is,

$$c_{AG} = \begin{cases} b_{AG}\pi^*_{Sij} & \text{for } I = 1 \\ b_{AG}(1 - \pi^*_{Sij}) & \text{for } I = 0 \end{cases} \tag{22}$$

Thus,

[4] Although it would perhaps have been more intuitively appealing to have $c_{BG}\ln(K_{i(j+1)} + 1)$ as a separate term, the maths in this case is horrendous. On the other hand, including c_{BG} within a single log term results in comparatively much simpler maths.

$$K_{i(j+1)I} = \begin{cases} \dfrac{b_{AG}\pi^*_{Sij}}{c_{TG}\pi^*_{Tij1} + c_{KG}} - \dfrac{1}{\pi_{Aij1} + c_{BG}} & \text{for } I = 1 \\ \dfrac{b_{AG}(1-\pi^*_{Sij})}{c_{TG}\pi^*_{Tij0} + c_{KG}} - \dfrac{1}{\pi_{Aij0} + c_{BG}} & \text{for } I = 0 \end{cases} \tag{23}$$

From Equation (23), if $K_{i(j+1)I} - K_{ij} \leq 0$ for either $I = 1$ or $I = 0$, then agent i has more connections than they would like. However, agent i cannot remove a fraction of a connection so will only consider severing a connection when this difference reaches 1. Thus, whenever $K_{i(j+1)I} - K_{ij} \leq -1$ and at least one connection acted toxically with the required polarity, then the agent will sever a connection. A single toxic sharer is selected randomly from all who shared toxically in the appropriate polarity, with equal probability per eligible connection.

7.6 Step 6

As for disconnections, agents decide whether to connect with new agents as a function of their optimal number of connections for each value of I, $K_{i(j+1)I}$, with the exception that agents who are disinterested do not seek new connections. Specifically, if $K_{ijI} - K_{ij} \geq 1$ for either $I = 1$ or $I = 0$ then agents will connect with a random agent who was an originator agent who shared with the same polarity as agent i in the same sharing chain (i.e., an agent selected from ω_{ijI} for round j, in the same sharing chain, with I matching that of the relevant agent). If no such agent exists, then agent i does not add a new connection in this step.

7.7 Step 7

Agents who have no connections after Step 6 connect with a random agent who's opinion is on the same side of 0.5 as themselves or whose opinion is within γ–a parameter representing tolerance for opposing views among those who are relatively undecided. For example, if $\gamma = 0.2$, then an agent with $\pi^*_{Sij} = 0.6$ would be willing to connect in an agent with $\pi^*_{Sij} \geq 0.4$.

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